

# Generative Adversarial Network Performance for Low Dimensional Data

Felix Jimenez<sup>1,2</sup>, Amanda Koepke<sup>2</sup>, Michael Frey<sup>2</sup>

<sup>1</sup>University of Colorado Boulder <sup>2</sup>National Institute of Standards and Technology

QPRC, 2019

# Outline

- 1 Motivating Problem
- 2 Introduction to GANs
- 3 Our Problem
  - 1D Distributions
  - 2D Distributions
- 4 Summary

# Motivating Problem

## The Problem

- Scientist makes measurements
- Needs to perform an analysis that requires more data
- Must use a method to generate more data using what is given

## The Problem Restated

- Given a sample  $\mathbf{x}_1, \dots, \mathbf{x}_n \sim p^*(\mathbf{x}), \mathbf{x}_i \in \mathcal{X}$
- We want a model,  $G_\theta$ , that can generate more samples from this distribution

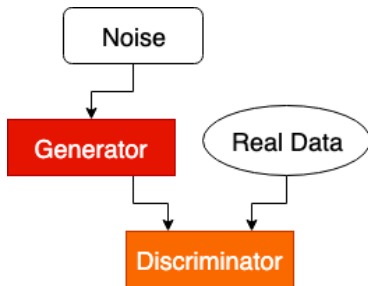
# GAN Basics

$$\mathbf{x}_1, \dots, \mathbf{x}_n \sim p^*(\mathbf{x}), \mathbf{x}_i \in \mathcal{X}$$

## Model

A Generative Adversarial Network (GAN) is made of a generator  $G_\theta(z)$ ,  
and a discriminator  $D_\phi(x)$

$$G_\theta(z) \mapsto \mathcal{X}, z \sim p(z) \quad D_\phi(x) \mapsto [0, 1]$$



GAN Diagram

## Loss

$$L(\theta, \phi) = E_{p(z)}[f(D_\phi(G_\theta(z)))] + E_{p^*(x)}[f(-D_\phi(x))]$$

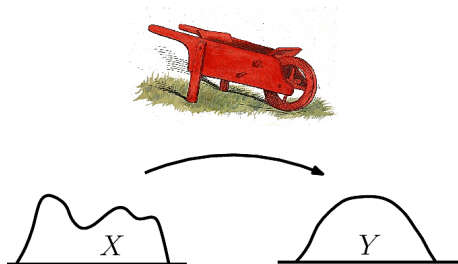
## Training

$$\min_{G_\theta} \max_{D_\phi} L(\theta, \phi)$$

## Our Choices

- Five Discriminator (critic) Updates Per Generator Update
- $f(x) = x$  (EM Distance)

# Earth Movers Distance

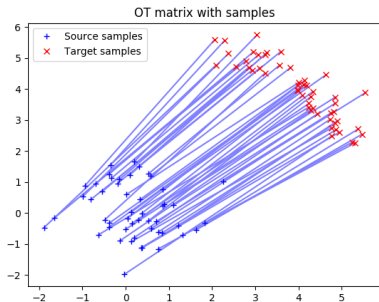


Transforming the distribution of  $X$  into the distribution of  $Y$

## EM Distance

$$\inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|]$$

# EM Distance Example



Optimal Transport Plan for Two Point Clouds

# The Problem We Are Working on

The questions we want to answer:

- How does the EM distance between the GAN samples and the true distribution change with training size?
- Is this relationship different for more complex distributions?

In this talk we focus on 1D and 2D examples.



# Implementation Details

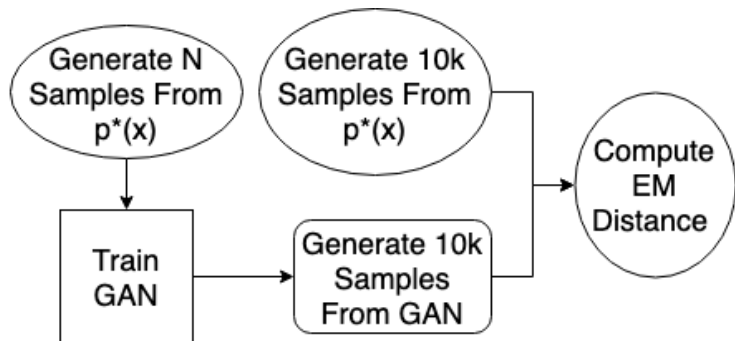


Diagram of Training and Evaluation Process

# Implementation Details Cont.

## Generator

| Layer           | Number of Units | Activation |
|-----------------|-----------------|------------|
| Fully Connected | 128             | Leaky-Relu |
| Fully Connected | 128             | Leaky-Relu |
| Fully Connected | 128             | Leaky-Relu |
| Fully Connected | Dim. of Output  | None       |

\*Generator input is a noise vector of length 128

## Discriminator

| Layer           | Number of Units | Activation |
|-----------------|-----------------|------------|
| Fully Connected | 32              | Leaky-Relu |
| Fully Connected | 32              | Leaky-Relu |
| Fully Connected | 32              | Leaky-Relu |
| Fully Connected | 1               | Leaky-Relu |

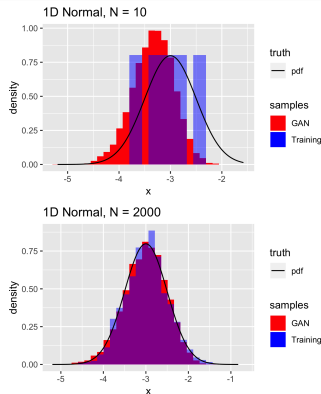
# Outline

- 1 Motivating Problem
- 2 Introduction to GANs
- 3 Our Problem**
  - 1D Distributions
  - 2D Distributions
- 4 Summary

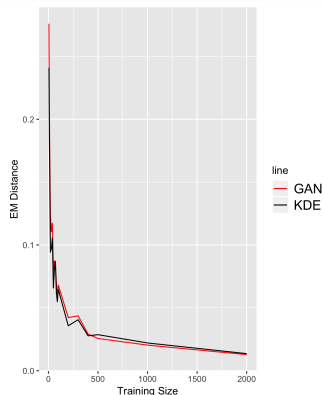
# 1D Normal Distribution

## True Distribution

$$x_1, \dots, x_n \sim N(\mu = -3, \sigma = 0.5)$$



Examples of GAN and Training Samples



EM Distance Between Truth and GAN vs. Training Size

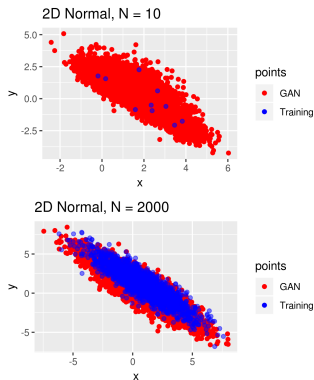
# Outline

- 1 Motivating Problem
- 2 Introduction to GANs
- 3 Our Problem**
  - 1D Distributions
  - **2D Distributions**
- 4 Summary

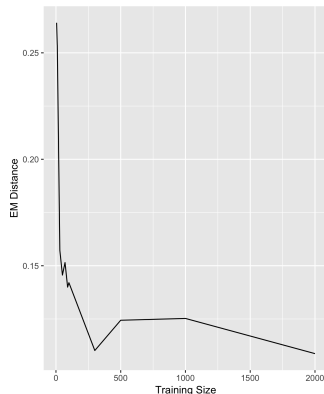
# Results for 2D Normal

## True Distribution

MVN with mean **1**, std of 2.125 and correlation of -0.9



Examples of GAN and Training Samples

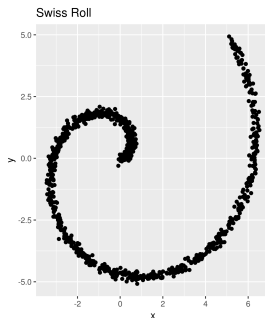


EM Distance Between Truth and GAN vs. Training Size

# Swiss Roll Distribution

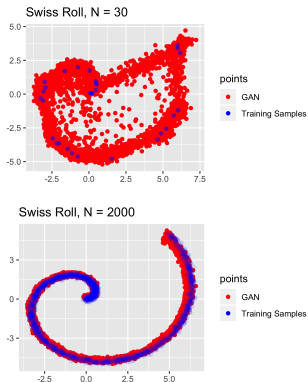
## Simulating Swiss Roll Samples

- $u \sim \text{unif}(0, 9\pi/4)$
- $w \sim \text{MVN}(\mu = \mathbf{0}, \Sigma = 0.1 * I)$
- $[x, y] = [u * \cos(u), u * \sin(u)] + w$

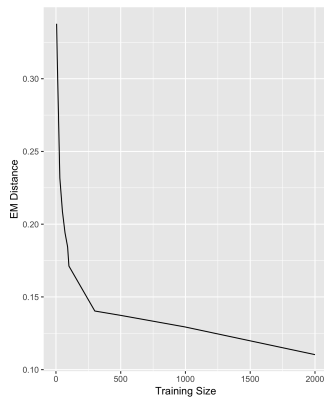


Realizations from Swiss Roll with Noise of 0.1

# Results for Swiss Roll



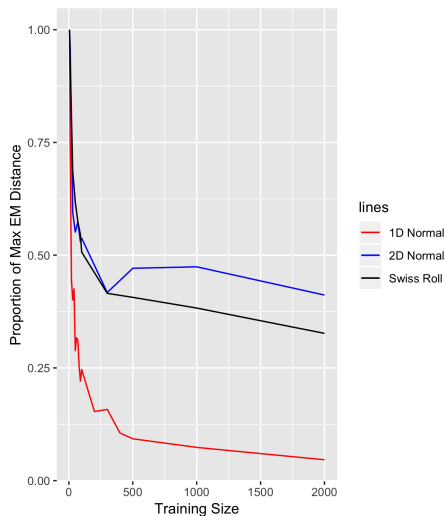
Examples of GAN and Training Samples



EM Distance Between Truth and GAN vs. Training Size



# Comparing Different EM Distances



Scaled EM Distance vs. Training Size for Different Distributions

# Summary

## Topics Covered

- EM Distance vs. Training
- Complexity of Distribution and EM Distance

## Next Steps

- Perform same analysis for more complex distributions
- Quantify "complexity" and examine relationship between EM Distance and Complexity